

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SOME BASIC CONCEPT OF CHEMISTRY

LECTURE-6

CHEMISTRY

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Baaz
Batch
Zindabad



JK Bole Wallah

Topics To Covered

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MOLE CONCEPT AND NUMERICALS

Yahan se Sawal Pakka
hai

REVISION.

1) Relative atomic mass.

$$= \frac{\text{Mass of an atom}}{\frac{1}{12} \text{ m of mass of an atom of } ^{12}\text{C}}$$

↓

$$= 1 \text{ amu}$$

$$= 1.66 \times 10^{-24} \text{ g.}$$

★ 2) Average atomic mass.

- Isotopes
- Relative abundance
+
Relative atomic mass

3) Formula mass. ④ Molecular mass.

- Ionic compounds.

$$\text{NaCl} = 23 + 35.5$$

$$=$$

MOLE CONCEPT AND NUMERICALS



The word "mole" was given by W. Ostwald & it means heap or pile.

It is defined as. that amount of substance having as many particle (atoms, or molecules or ions etc) as there are atoms in 12 g of ^{12}C . (i.e. 6.022×10^{23})

This number 6.022×10^{23} is also called Avagadro number & is denoted by " N_A ".

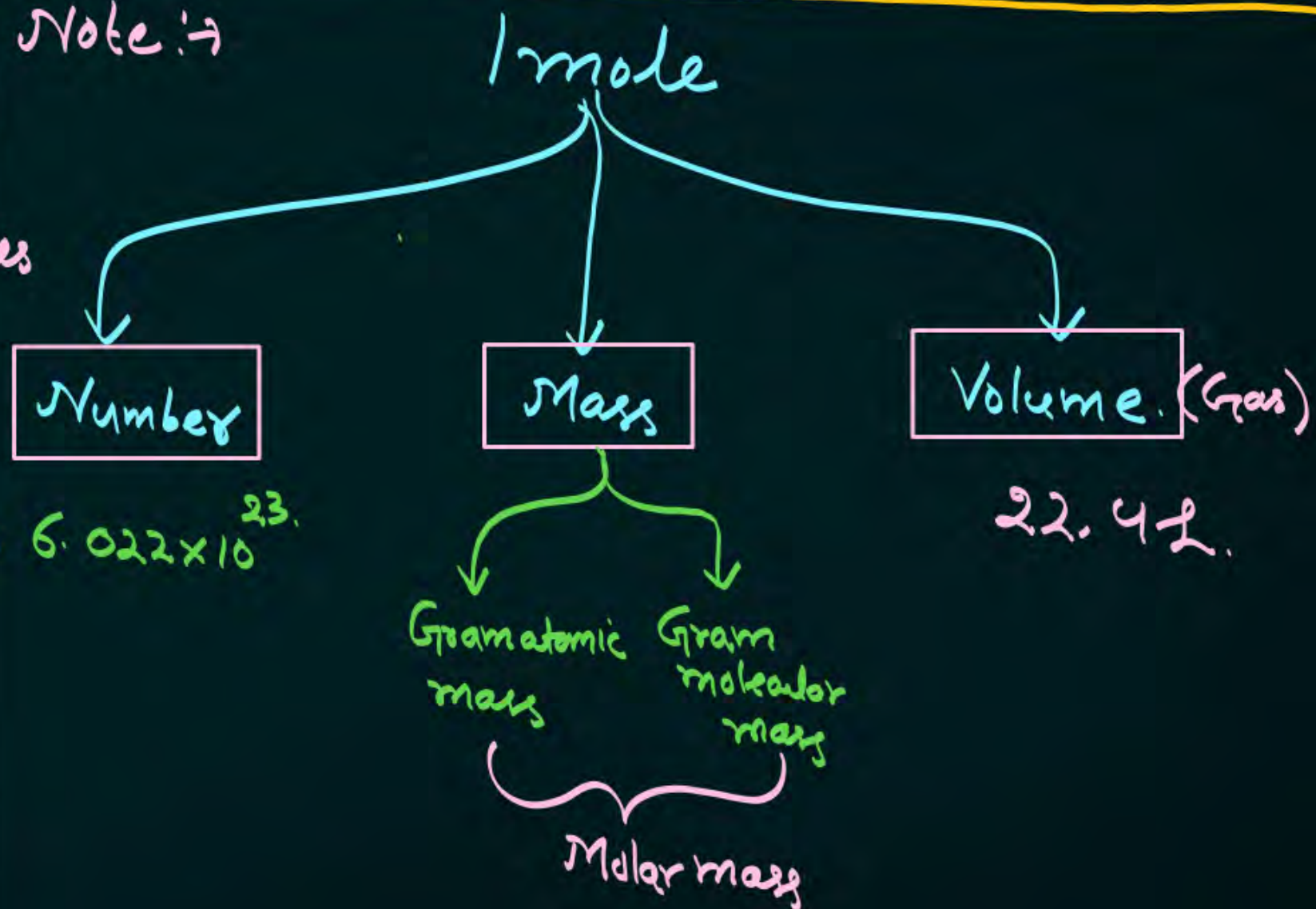
i.e. 1 mole of any atom = 6.022×10^{23} atoms.

1 mole of any molecule = 6.022×10^{23} molecule.

1 mole of electron = 6.022×10^{23} electron.

1 mole of Vishal Sir = 6.022×10^{23} Vishal Sir.

Note: $\rightarrow$



1 mole

Sample	No. of particles	Molar mass $\rightarrow$ (Mass of one mole of any substance)	Volume
Na(s)	6.022×10^{23} atoms of Na.	23 g/mol	X
H ₂ O(g)	6.022×10^{23} molecules of H ₂ O.	18 g/mol.	22.4 L.
CO ₂ (g)	6.022×10^{23} molecules of CO ₂ .	44 g/mol.	22.4 L.
Li	6.022×10^{23} atom of Li.	7 g/mol.	X

Some Important Formula



1. No. of moles of atom = $\frac{\text{Given atomic mass}}{\text{Atomic mass}}$
2. No. of moles of molecule = $\frac{\text{Given molecular mass}}{\text{Molecular mass}}$
3. No. of moles of a Gas = $\frac{\text{Given volume}}{22.4 \text{ L}}$
4. $\star$ No. of molecules = No. of moles $\times N_A$
5. $\star$ No. of atoms = No. of molecules $\times$ Atomicity
- 6) Mass of an atom = $\frac{\text{Atomic mass}}{6.022 \times 10^{23}}$
- 7) Mass of one molecule = $\frac{\text{Molecular mass}}{6.022 \times 10^{23}}$

Atomicity is the no. of atoms present in a molecule

e.g. $\text{NaCl} = 2$

$\text{CO}_2 = 3$

$\text{H}_2\text{O} = 3$

$\text{H}_2\text{SO}_4 = 7$

QUESTION



Calculate the number of atoms in each of the following

(i) 52 moles of Ar

(ii) 52 g of He.

i) Solution $\Rightarrow$

$$\begin{aligned}\text{No. of atoms} &= \text{No. of molecules} \times \text{Atomicity} \\ &= \text{No. of moles} \times N_A \times \text{Atomicity} \\ &= 52 \times N_A \times 1 \\ &= \underline{52 N_A}\end{aligned}$$

$$\text{He} = 4g$$

$$\begin{aligned}\text{ii) No. of atoms} &= \text{No. of molecules} \times \text{Atomicity} \\ &= \text{No. of moles} \times N_A \times \text{Atomicity} \\ &= \frac{\text{Given atomic mass}}{\text{Atomic mass of He}} \times N_A \times 1 \\ &= \frac{52}{4} N_A \\ &= 13 N_A\end{aligned}$$

QUESTION

Community Me. Post
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Home Work

Which one of the following will have the largest number of atoms?

(i) 1 g Au(s) ($A_u = 107$)

(ii) 1 g Na(s)

(iii) 1 g Li(s)

(iv) 1 g of $Cl_2(g)$

$$\begin{aligned} \text{i) No. of atoms} &= \text{No. of molecules} \times \text{Atomicity} \\ &= \text{No. of moles} \times N_A \times \text{Atomicity} \\ &= \frac{\text{Given mass of Au}}{\text{Atomic mass of Au}} \times N_A \times \text{Atomicity} \\ &= \frac{1}{107} \times N_A \end{aligned}$$

QUESTION



$$C + O_2 = 12 + 32 = 44$$

$$\text{No. of molecules} = \text{No. of moles} \times N_A$$

Which of the following has maximum number of molecules

(a) 44 gram of carbon dioxide

(b) 48 gram of O_2

(c) 8 gram of H_2

a) Sol.

$$\begin{aligned}\text{No. of molecules} &= \text{No. of moles} \times N_A \\ &= \frac{\text{Given mass}}{\text{Molecular mass}} \times N_A \\ &= \frac{44}{44} \times N_A \\ &= \boxed{1 N_A} \text{ or } 6.022 \times 10^{23}\end{aligned}$$

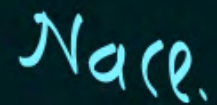
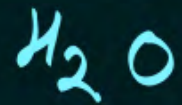
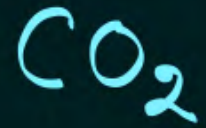
$$\begin{aligned}\text{b) Sol. No. of molecules} &= \text{No. of moles} \times N_A \\ &= \frac{\text{Given mass}}{\text{Molecular mass of } O_2} \times N_A \\ &= \frac{48}{32} \times N_A \\ &= \boxed{1.5 N_A}\end{aligned}$$

c) Sol.

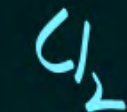
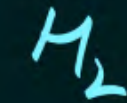
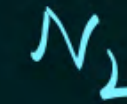
$$\begin{aligned}\text{No. of molecules} &= \text{No. of moles} \times N_A \\ &= \frac{\text{Given mass of } H_2}{\text{Molecular mass of } H_2} \times N_A = \frac{8}{2} \times N_A \\ &= \boxed{4 N_A}\end{aligned}$$

① All compounds are molecules but not all molecules are compounds.

↓
different atoms



↓



QUESTION



Q. Calculate the number of moles in 9 gram of water. (H_2O)

$$\text{Sol:} \rightarrow \text{No. of mole} = \frac{\text{Given mass of } H_2O}{\text{Molecular mass of } H_2O} = \frac{9}{18} = \frac{1}{2} \text{ mole.}$$

QUESTION



Which of the following has maximum number of molecules

(a) 15 l of H_2 gas

(b) 5 l of N_2 gas ✓

a) Sol: $\rightarrow$ No. of molecules = No. of moles $\times N_A$

$$= \frac{\text{Given Volume}}{22.4 \text{ l}} \times N_A$$
$$= \frac{15}{22.4} \times N_A$$
$$= 0.66 \times N_A$$

b) No. of molecule = No. of mole $\times N_A$

$$= \frac{\text{Given Volume}}{22.4 \text{ l}} \times N_A$$
$$= \frac{5}{22.4} \times N_A$$
$$= 0.22 N_A$$

QUESTION

1 mole of ethane = C_2H_6



In three moles of ethane (C_2H_6), calculate the following

- (i) Number of moles of carbon atoms.
- (ii) Number of moles of hydrogen atoms.
- (iii) Number of molecules of ethane.

i) Ans One mole of ethane has = 2-mole of C
three mole of ethane = $3 \times 2 = 6$ mole of C

2) Ans.

one mole of ethane molecule = 6 mole of H
3 mole of ethane = 3×6
= 18 mole of H.

3) No. of molecules = No. of moles $\times N_A$
= $3 \times N_A$.

QUESTION



How many molecules and atoms of sulphur are present in 0.1 mole of S₈ molecule? → Atomicity = 8

Sol.

i) No. of molecules.

$$= \text{No. of moles} \times N_A$$

$$= 0.1 \times N_A$$

ii) No. of atoms.

$$= \text{No. of molecules} \times \text{Atomicity}$$

$$= \text{No. of moles} \times N_A \times \text{Atomicity}$$

$$= 0.1 \times N_A \times 8$$

$$= 0.8 N_A$$

QUESTION



What will be the mass of one ^{12}C atom in g?

Sol. Mass of an atom = $\frac{\text{Atomic mass}}{N_A} = \frac{12}{6.022 \times 10^{23}} = 1.99 \times 10^{-23}$

Q. Calculate the mass of one CO_2 molecule in g? = $\frac{1.99}{10^{23}}$

Mass of a molecule = $\frac{\text{Molecular mass}}{\text{Avogadro no.}}$

= $\frac{44}{6.022 \times 10^{23}}$



PERCENTAGE COMPOSITION

It gives the number of grams of the element present in 100 *g* of the compound. It is obtained by dividing the mass of each element in 1 mole of the compound by molar mass of the compound and multiplying by 100. Thus,

$$= \frac{\text{Mass of that element in 1 mole compound}}{\text{Molar mass of component}} \times 100$$

PERCENTAGE COMPOSITION

It is the mass of an element present in 100 gm of compound.

$$\text{Mathematically} = \frac{\text{Mass of an element in compound}}{\text{Total mass of compound}} \times 100$$

For Example. We can calculate the percentage composition of Hydrogen & Oxygen in H_2O as.

i) Percentage composition of Hydrogen in H_2O .

$$= \frac{2}{18} \times 100$$
$$= 11\%$$

ii) Percentage composition of Oxygen in H_2O .

$$= \frac{16}{18} \times 100 = 89\%$$
$$= 0.89 \times 100$$



Q. What is % composition of C + oxygen in CO_2 ?

VISHAL BHAIYA KI
REQUEST.

PREVIOUS LECTURE DEKH K Aaya.
KARO

Vishal Sir.

Thank
You

✓ 1 amu or u

✓ 1 g/atom

✓ 1 g/mol